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# Expected Returns in Treasury Bonds

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Photo by Stefan Els

# Introduction

- **Purpose:** forecast Treasury bond returns
- A variation of a framework developed by **Cieslak and Povala** in 2011
- **Goal:** apply the framework to SA context and use it for investment decisions

- Cochrane and Piazzesi (2005)
- Cieslak and Povala (2011)
- Adrian and Moench (2013)

## INFLATION

- Monthly US and SA Core CPI
- Time period:
  - US: 1966 to 2024
  - SA: 1980 to 2024
- **Core CPI inflation:** year-on-year simple growth rate in the Core CPI index

## TRADED BOND DATA

- Convexity, modified duration and traded bond yields across different maturities.
- Time period:
  - US: 1962 to 2024
  - SA: 1995 to 2024

## ZERO-COUPON YIELDS

- Monthly zero-coupon yields for bonds maturing in 1, 2, 5, 7, 10, and 20 years in US and SA
- Time period:
  - US: 1969 to 2024
  - SA: 1995 to 2024

# Summary

**Step 1: Generate cycles**

**Step 2: Use the cycles to predict future bond returns**

# Methodology – Step 1

The yield at time  $t$  of a zero-coupon bond with  $n$  years to maturity:

$$y_t^{(n)} = b_0^{(n)} + b_\tau^{(n)}\tau_t + b_r^{(n)}r_t + b_\pi^{(n)}\pi_t^c + rpy_t^{(n)}$$

where

- $\tau_t \equiv$  trend inflation
- $\pi_t^c \equiv$  transitory inflation
- $r_t \equiv$  real rate
- $rpy_t^{(n)} \equiv$  risk premium

**Step 1: Generate cycles**

# Methodology – Step 1

The yield at time  $t$  of a zero-coupon bond with  $n$  years to maturity:

$$y_t^{(n)} = b_0^{(n)} + b_\tau^{(n)} \tau_t + \boxed{b_r^{(n)} r_t + b_\pi^c^{(n)} \pi_t^c + rpy_t^{(n)}}$$

where

transitory portion

- $\tau_t \equiv$  trend inflation
- $\pi_t^c \equiv$  transitory inflation
- $r_t \equiv$  real rate
- $rpy_t^{(n)} \equiv$  risk premium

**Step 1: Generate cycles**

# Methodology – Step 1

zero-coupon bond yield

$$y_t^{(n)} = b_0^{(n)} + b_\tau^{(n)} \tau_t + b_r^{(n)} r_t + b_\pi^{(n)} \pi_t^c + rpy_t^{(n)}$$

$$y_t^{(n)} = b_0^{(n)} + b_\tau^{(n)} \tau_t + \tilde{c}_t^{(n)}$$

**Step 1: Generate cycles**

# Methodology – Step 1

$$y_t^{(n)} = b_0^{(n)} + b_\tau^{(n)} \tau_t + \boxed{b_r^{(n)} r_t + b_\pi^{(n)} \pi_t^c + rpy_t^{(n)}}$$

$$\boxed{y_t^{(n)}} = b_0^{(n)} + b_\tau^{(n)} \boxed{\tau_t} + \boxed{\tilde{c}_t^{(n)}}$$

cointegrated

Step 1: Generate cycles

# Methodology – Step 1

zero-coupon bond yield

residuals  $\equiv$  cycles

$$y_t^{(n)} = b_0^{(n)} + b_\tau^{(n)} \tau_t^{proxy} + \epsilon_t^{(n)} \quad (\epsilon_t^{(n)} = c_t^{(n)})$$
$$\tau_t^{proxy} = \frac{\sum_{i=0}^{119} v^i CPI_{t-i}}{\sum_{i=0}^{119} v^i}$$

trend inflation

**Step 1: Generate cycles**

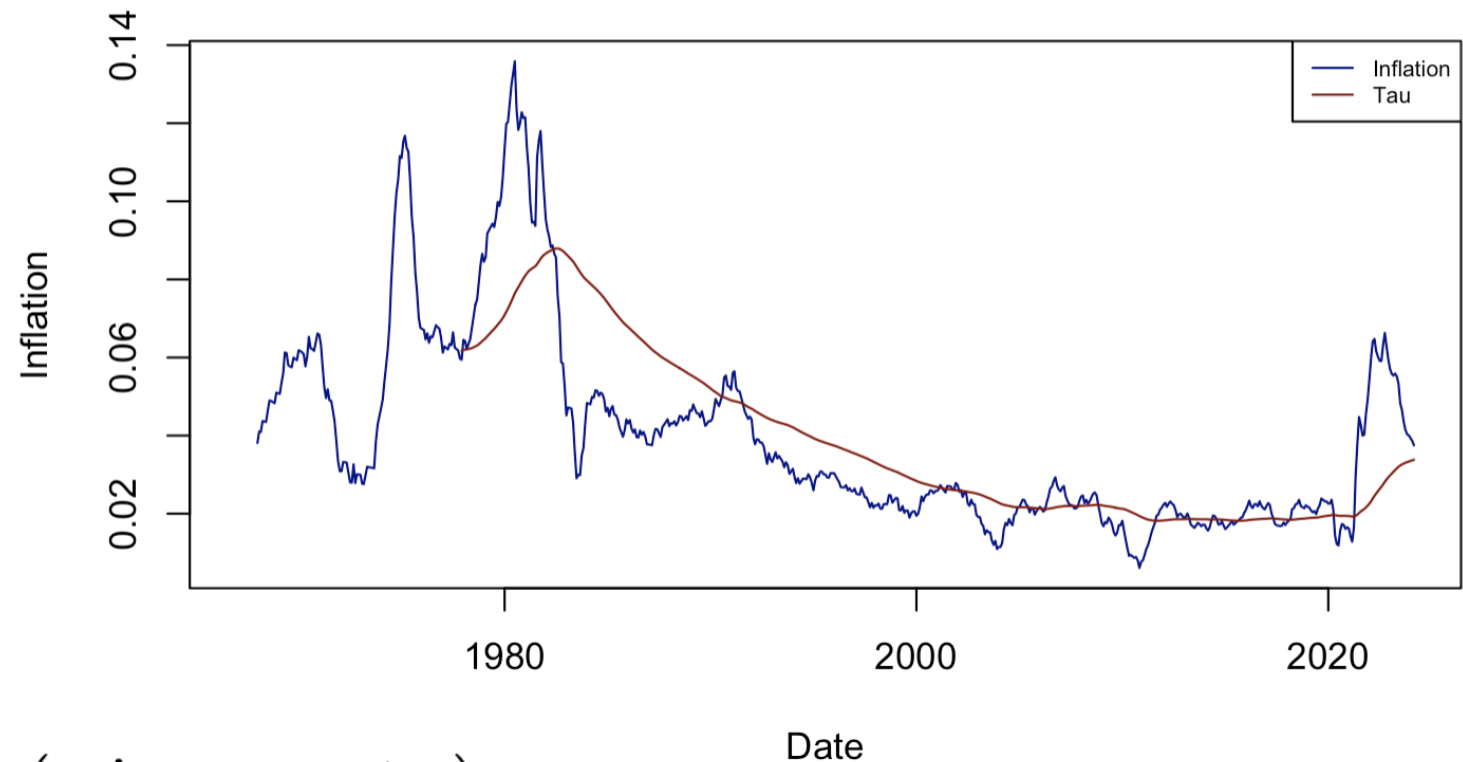
# Trend Inflation

A proxy for the trend inflation

$$\tau_t^{proxy} = \frac{\sum_{i=0}^{119} v^i CPI_{t-i}}{\sum_{i=0}^{119} v^i}$$

where

- $v = 0.987 \equiv$  discounting factor (gain parameter)



# Methodology – Returns

$$rx_{t+12}^{(n)} = p_{t+12}^{(n-1)} - p_t^{(n)} - y_t^{(1)}$$

where

- $p_t^{(n)} \equiv \log$  price of a zero-coupon bond with  $n$  years to maturity at time  $t$
- $y_t^{(1)}$  is the 1-year zero-coupon bond yield

\*Since monthly data is used,  $t+12$  in  $rx_{t+12}^{(n)}$  represents 12 months into the future, which is equivalent to one year ahead.

# Methodology – Returns

$$rp_{t+12}^{(n)} = -(\text{MODIFIED DURATION}_t)(y_{t+12}^{*(n)} - y_t^{*(n)}) + \frac{1}{2}(\text{CONVEXITY}_t)(y_{t+12}^{*(n)} - y_t^{*(n)})^2$$

where

- $y_t^{*(n)} \equiv$  yield of a traded bond with maturity  $n$  at time  $t$

# Methodology – Full Model

## Step 2: Use the cycles to predict future bond returns

$$\boxed{y_t^{(n)}} = b_0^{(n)} + b_{\tau}^{(n)} \tau_t^{proxy} + \boxed{\epsilon_t^{(n)}}$$

zero-coupon bond yield      residuals  $\equiv$  cycles       $\boxed{\epsilon_t^{(n)}} = \boxed{c_t^{(n)}}$

The **cycles** are used to forecast expected returns using this regression model:

$$rp_{t+12}^{(n)} = \delta_0 + \sum_i \delta_i \boxed{c_t^{(i)}} + \epsilon_{t+12}^{(n)}$$

where

- $i = \{1, 2, 5, 7, 10, 20\}$
- $rp_{t+12}^{(n)} \equiv$  12-month forward-looking traded bond return

# Methodology – Factor Model

**Step 2.1: Obtain the return forecasting factor,  $\hat{c}^f$**

$$\overline{r}p_{t+12} = \gamma_0 + \gamma_1 c_t^{(1)} + \gamma_2 \bar{c}_t + \varepsilon_{t+12}$$

where

- $\overline{r}p_{t+12}$  is the average yield of traded bonds across different maturities at time  $t+12$
- $c_t^{(1)}$   $\equiv$  residuals from regressing the zero coupon yield with maturity of 1 year on  $\tau_t$  at time  $t$ .
- $\bar{c}_t = \frac{1}{m-1} \sum_{i=2}^{20} c_t^{(i)} \equiv$  average of the residuals from regressing the zero coupon yields of different maturities (excluding a maturity of 1 year) on  $\tau_t$  at time  $t$ .

# Methodology – Factor Model

**Step 2.2: Use  $\hat{cf}$  to predict future bond returns**

return forecasting factor

$$\widehat{cf}_t = \hat{\gamma}_0 + \hat{\gamma}_1 c_t^{(1)} + \hat{\gamma}_2 \bar{c}_t$$

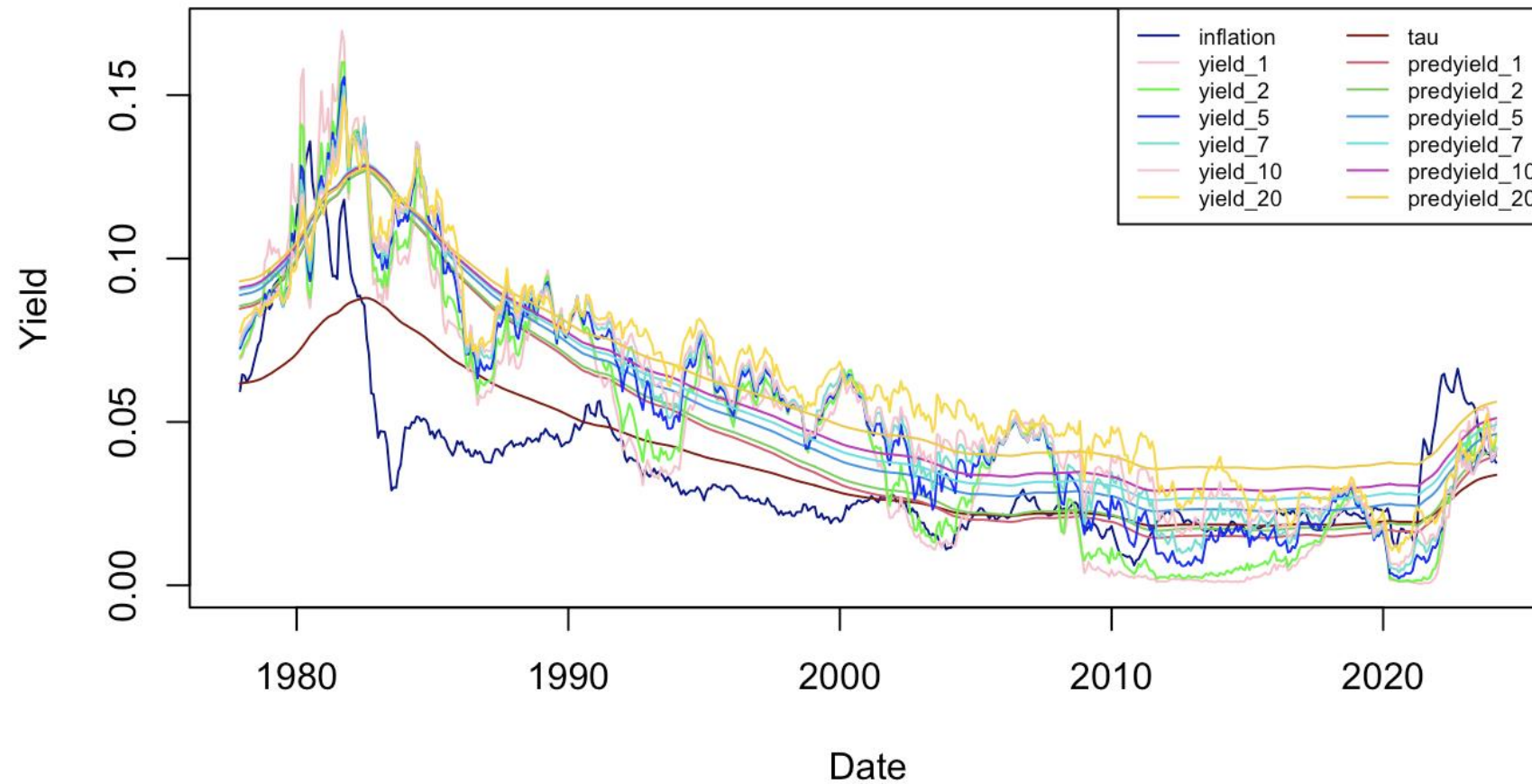
$$rp_{t+12}^{(n)} = \beta_0^{(n)} + \beta_1^{(n)} \widehat{cf}_t + \varepsilon_{t+12}^{(n)}$$

# Results

US Context

$$y_t^{(n)} = b_0^{(n)} + b_\tau^{(n)} \tau_t^{proxy} + \epsilon_t^{(n)}$$

zero-coupon bond yield



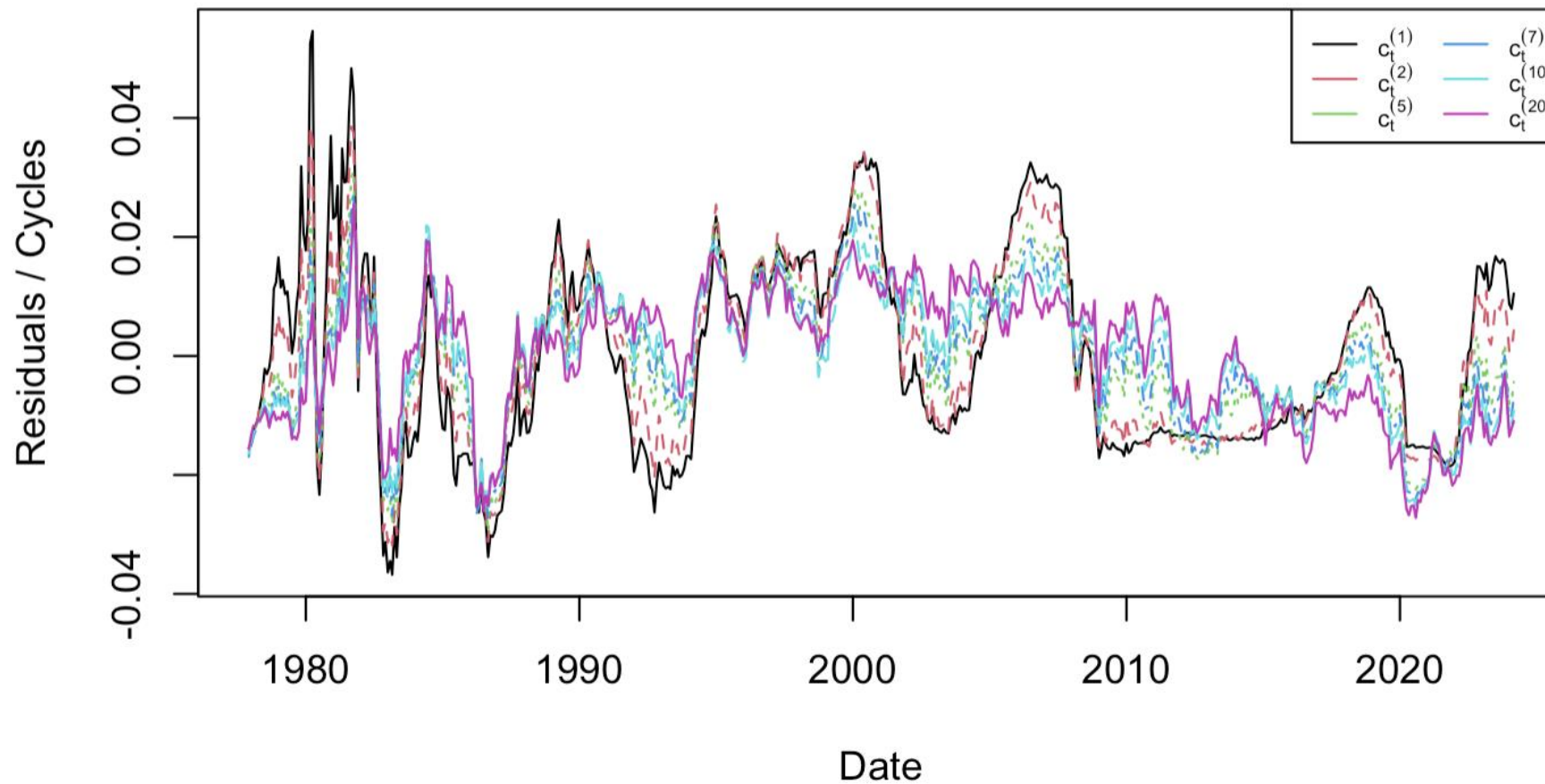
# Results

US Context

$$y_t^{(n)} = b_0^{(n)} + b_\tau^{(n)} \tau_t^{proxy} + \epsilon_t^{(n)}$$

zero-coupon bond yield

cycles



# Results

## US Context

Table 1: p-values from ADF test for  $\tau_t^{CPI}$  and zero-coupon yields

| No. of lags | $\tau_t^{CPI}$ | $y_t^{(1)}$ | $y_t^{(2)}$ | $y_t^{(5)}$ | $y_t^{(7)}$ | $y_t^{(10)}$ | $y_t^{(20)}$ |
|-------------|----------------|-------------|-------------|-------------|-------------|--------------|--------------|
| 1           | 0.01961        | 0.1591      | 0.1869      | 0.1074      | 0.1093      | 0.0936       | 0.0545       |
| 2           | 0.5088         | 0.3659      | 0.3811      | 0.2529      | 0.2178      | 0.1857       | 0.1087       |
|             | > 0.05         | > 0.05      | > 0.05      | > 0.05      | > 0.05      | > 0.05       | > 0.05       |

Table 2: p-values from ADF test for cointegrating residual

| No. of lags | $c_t^{(1)}$ | $c_t^{(2)}$ | $c_t^{(5)}$ | $c_t^{(7)}$ | $c_t^{(10)}$ | $c_t^{(20)}$ |
|-------------|-------------|-------------|-------------|-------------|--------------|--------------|
| 1           | < 0.01      | < 0.01      | < 0.01      | < 0.01      | < 0.01       | < 0.01       |
|             | < 0.05      | < 0.05      | < 0.05      | < 0.05      | < 0.05       | < 0.05       |

# Results

## US Context

$$\text{traded bond return } rp_{t+12}^{(n)} = \delta_0 + \sum_i \delta_i c_t^{(i)} + \varepsilon_{t+12}^{(n)}$$

Table 3: In-sample adjusted  $R^2$  values for different maturities using OLS and Elastic Net regression in US context - full model

| $R^2$       | $rp^{(1)}$ | $rp^{(2)}$ | $rp^{(5)}$ | $rp^{(7)}$ | $rp^{(10)}$ | $rp^{(20)}$ |
|-------------|------------|------------|------------|------------|-------------|-------------|
| OLS         | 0.2555     | 0.2883     | 0.3530     | 0.3833     | 0.3928      | 0.3900      |
| Elastic Net | 0.2543     | 0.2868     | 0.3528     | 0.3528     | 0.3611      | 0.3823      |

# Results

## US Context

Table 4: Comparison of in-sample adjusted  $R^2$  values in US context

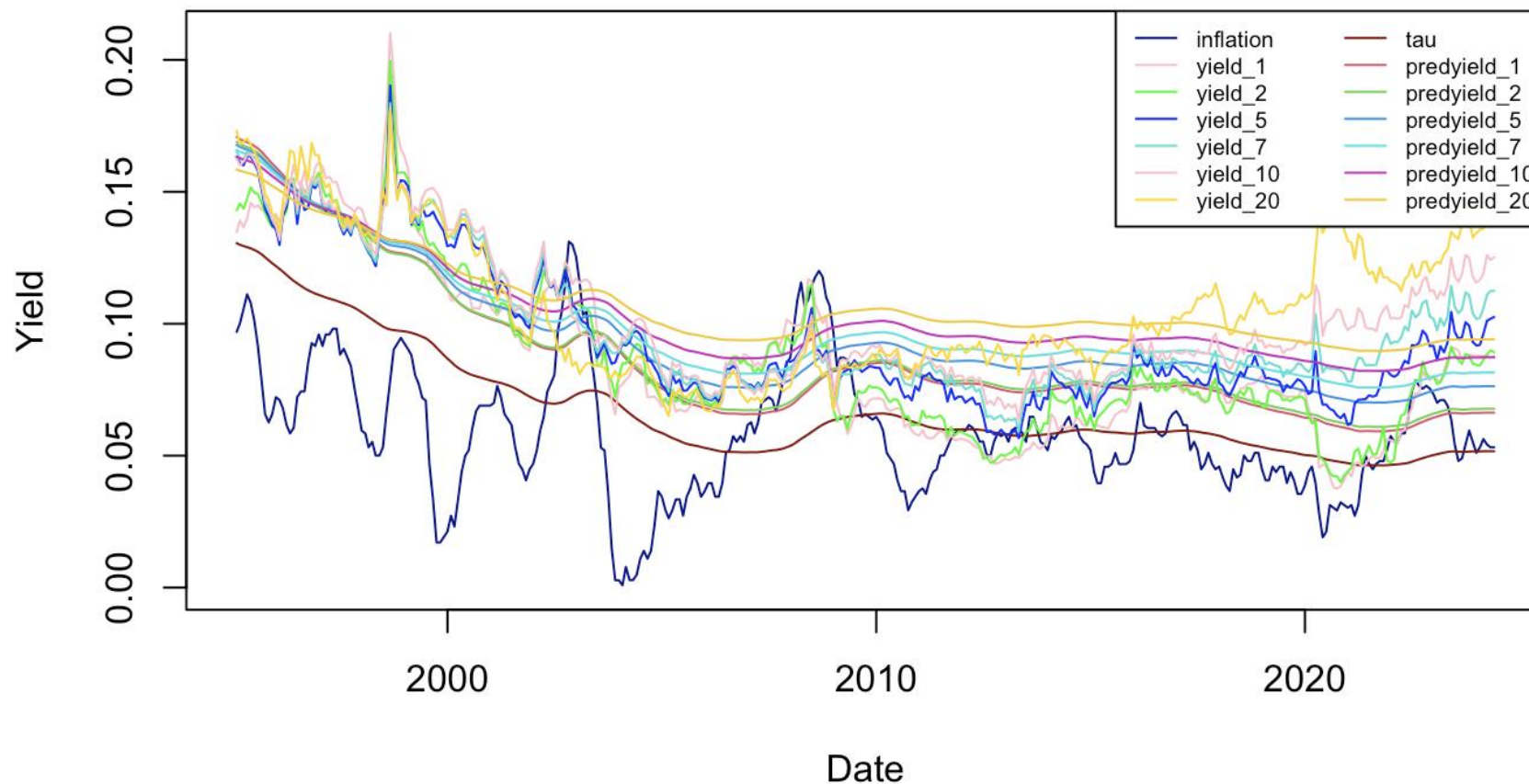
| $R^2$                    | $rp^{(1)}$ | $rp^{(2)}$ | $rp^{(5)}$ | $rp^{(7)}$ | $rp^{(10)}$ | $rp^{(20)}$ |
|--------------------------|------------|------------|------------|------------|-------------|-------------|
| OLS - full model         | 0.2555     | 0.2883     | 0.353      | 0.3833     | 0.3928      | 0.3900      |
| Elastic Net - full model | 0.2543     | 0.2868     | 0.3528     | 0.3528     | 0.3611      | 0.3823      |
| OLS - factor model       | 0.1733     | 0.2303     | 0.3142     | 0.3413     | 0.3505      | 0.3522      |

# Results

SA Context

$$y_t^{(n)} = b_0^{(n)} + b_\tau^{(n)} \tau_t^{proxy} + \epsilon_t^{(n)}$$

zero-coupon bond yield



# Results

SA Context

$$y_t^{(n)} = b_0^{(n)} + b_\tau^{(n)} \tau_t^{proxy} + \epsilon_t^{(n)}$$

zero-coupon bond yield

cycles

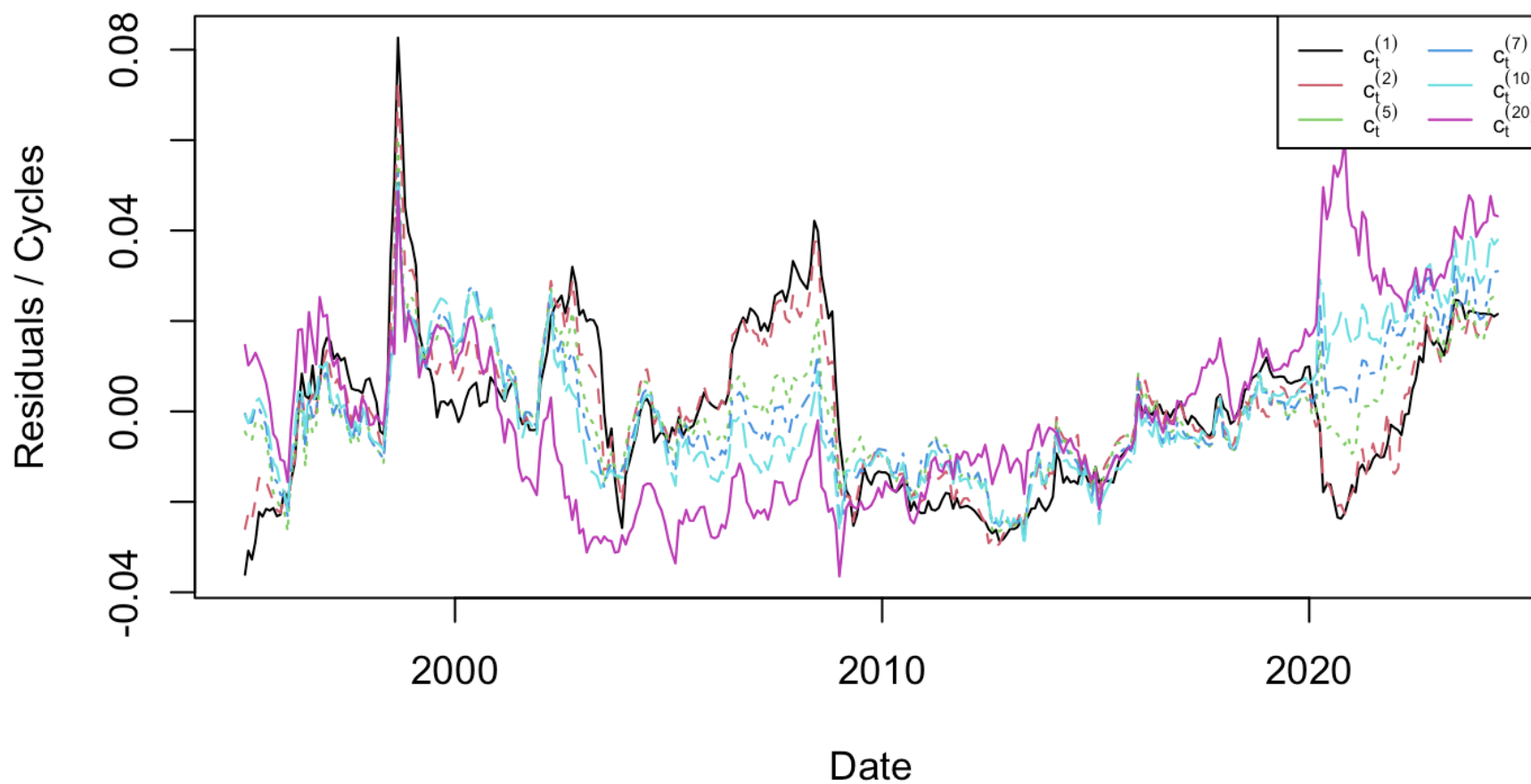


Table 5: p-values from ADF test for  $\tau_t^{CPI}$  and SA zero-coupon yields

| No. of lags | $\tau_t^{CPI}$ | $y_t^{(1)}$ | $y_t^{(2)}$ | $y_t^{(5)}$ | $y_t^{(7)}$ | $y_t^{(10)}$ | $y_t^{(20)}$ |
|-------------|----------------|-------------|-------------|-------------|-------------|--------------|--------------|
| 1           | 0.3509         | 0.3873      | 0.3890      | 0.5172      | 0.6068      | 0.7076       | 0.6388       |
|             | > 0.05         | > 0.05      | > 0.05      | > 0.05      | > 0.05      | > 0.05       | > 0.05       |

Table 6: p-values from ADF test for cointegrating residuals with SA data

| No. of lags | $c_t^{(1)}$ | $c_t^{(2)}$ | $c_t^{(5)}$    | $c_t^{(7)}$ | $c_t^{(10)}$ | $c_t^{(20)}$ |
|-------------|-------------|-------------|----------------|-------------|--------------|--------------|
| 1           | 0.0271      | 0.0215      | 0.0728         | 0.2226      | 0.4187       | 0.5219       |
|             | < 0.05      | < 0.05      | > 0.05         | > 0.05      | > 0.05       | > 0.05       |
|             | stationary  |             | non-stationary |             |              |              |

# Results

## SA Context

$$\text{traded bond return } rp_{t+12}^{(n)} = \delta_0 + \sum_i \delta_i c_t^{(i)} + \varepsilon_{t+12}^{(n)}$$

Table 7: In-sample adjusted  $R^2$  values for different maturities using OLS and Elastic Net regression in SA context - full model

| $R^2$       | $rp^{(1)}$ | $rp^{(2)}$ | $rp^{(5)}$ | $rp^{(7)}$ | $rp^{(10)}$ | $rp^{(20)}$ |
|-------------|------------|------------|------------|------------|-------------|-------------|
| OLS         | 0.3040     | 0.2838     | 0.3248     | 0.3241     | 0.3254      | 0.3195      |
| Elastic Net | 0.2731     | 0.2815     | 0.3228     | 0.3257     | 0.3250      | 0.3124      |

# Results

## SA Context

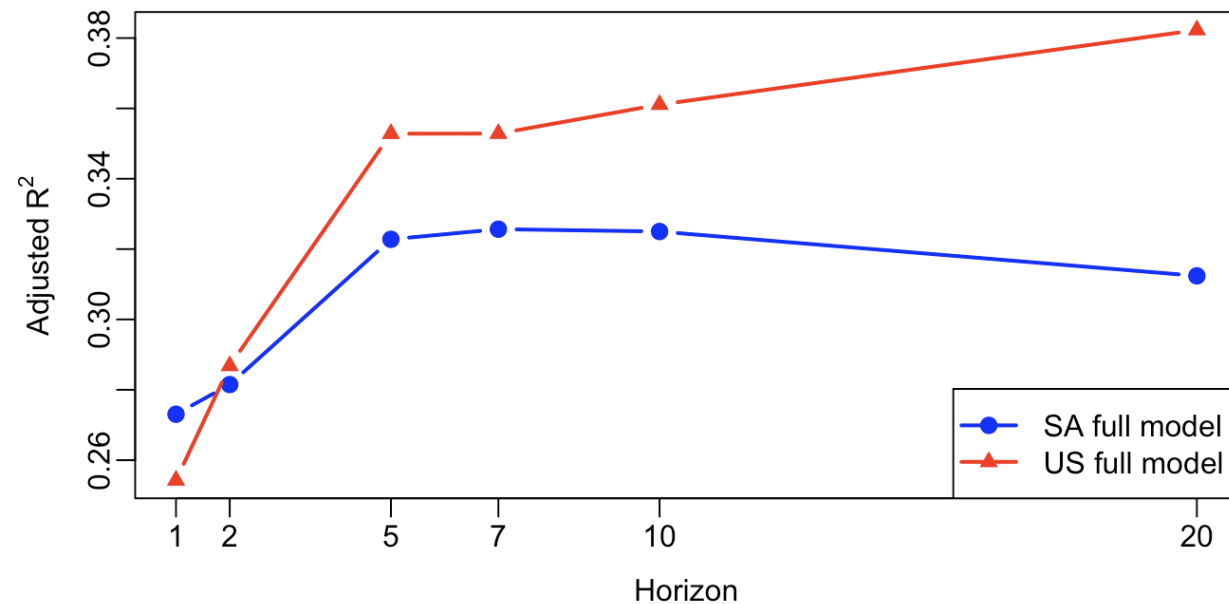
Table 8: Comparison of in-sample adjusted  $R^2$  values in SA context

| $R^2$                    | $rp^{(1)}$ | $rp^{(2)}$ | $rp^{(5)}$ | $rp^{(7)}$ | $rp^{(10)}$ | $rp^{(20)}$ |
|--------------------------|------------|------------|------------|------------|-------------|-------------|
| OLS - full model         | 0.3040     | 0.2838     | 0.3248     | 0.3241     | 0.3254      | 0.3195      |
| Elastic Net - full model | 0.2731     | 0.2815     | 0.3228     | 0.3257     | 0.3250      | 0.3124      |
| OLS - factor model       | 0.1304     | 0.1556     | 0.1246     | 0.1221     | 0.0961      | 0.1092      |

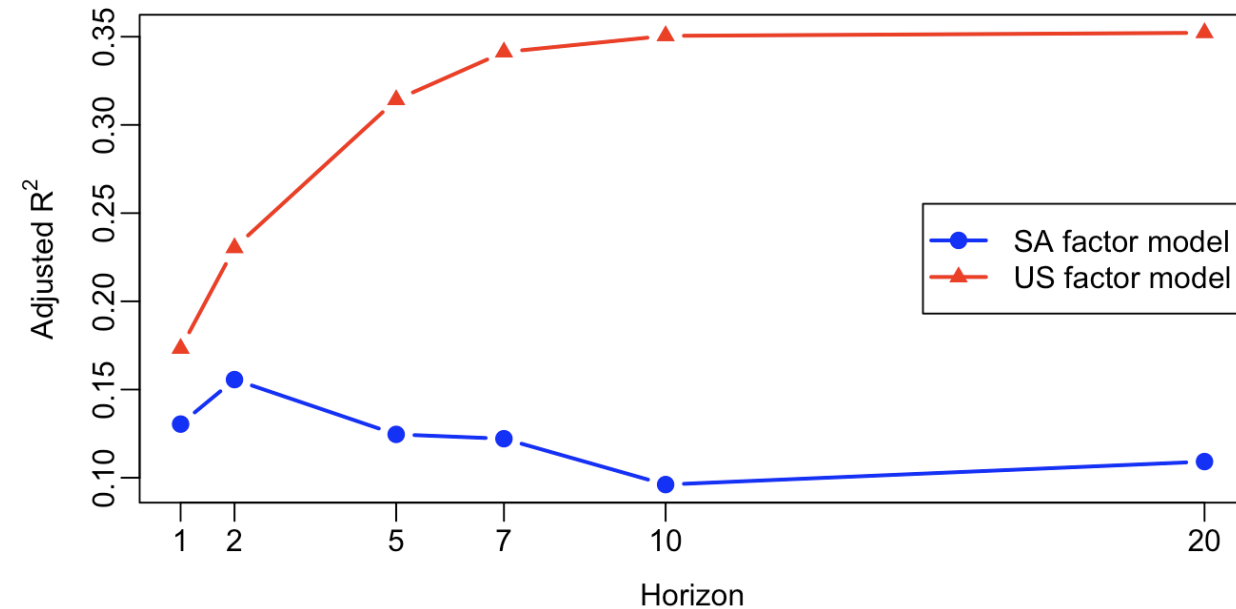
# Results

## US & SA Comparison

**Biplot of Full Model Results for SA and US Using Elastic Net**



**Biplot of Factor Model Results for SA and US**



## Step 2: Use the cycles to predict future bond returns

$$rp_{t+12}^{(n)} = \delta_0 + \sum_i \delta_i c_t^{(i)} + \sum_i \delta_i^* c_t^{*(i)} + \varepsilon_{t+12}^{(n)}$$

where

- $rp_{t+12}^{(n)} \equiv$  SA traded bond returns
- $c_t^{(i)} \equiv$  SA cycles
- $c_t^{*(i)} \equiv$  US cycles

# Results

## Combining US and SA Cycles

$$rp_{t+12}^{(n)} = \delta_0 + \sum_i \delta_i c_t^{(i)} + \sum_i \delta_i^* c_t^{*(i)} + \varepsilon_{t+12}^{(n)}$$

traded bond return

Table 9: In-sample adjusted  $R^2$  values obtained from combining the US and SA cycles using OLS and Elastic Net regression

| $R^2$       | $rp^{(1)}$ | $rp^{(2)}$ | $rp^{(5)}$ | $rp^{(7)}$ | $rp^{(10)}$ | $rp^{(20)}$ |
|-------------|------------|------------|------------|------------|-------------|-------------|
| OLS         | 0.5034     | 0.4681     | 0.4828     | 0.5097     | 0.5370      | 0.5798      |
| Elastic Net | 0.5034     | 0.4677     | 0.4812     | 0.5119     | 0.5329      | 0.5727      |

# Results

Out-of-sample –  $R^2$

Table 10: Comparison of out-of-sample  $R^2$  values in US, SA and SA with SA & US cycles

| $R^2$                             | $rp^{(1)}$ | $rp^{(2)}$ | $rp^{(5)}$ | $rp^{(7)}$ | $rp^{(10)}$ | $rp^{(20)}$ |
|-----------------------------------|------------|------------|------------|------------|-------------|-------------|
| <b>US</b>                         |            |            |            |            |             |             |
| OLS - full model                  | -0.2219    | -0.2226    | -0.1922    | -0.1373    | -0.1272     | -0.1668     |
| Elastic Net - full model          | -0.1582    | -0.1537    | -0.1423    | -0.0971    | -0.1116     | -0.1299     |
| OLS - factor model                | -0.1647    | -0.1510    | -0.1310    | -0.0948    | -0.0945     | -0.0807     |
| <b>SA</b>                         |            |            |            |            |             |             |
| OLS - full model                  | -0.6066    | -0.4438    | -0.5451    | -0.582     | -0.7594     | -1.1556     |
| Elastic Net - full model          | -0.7523    | -0.4079    | -0.5298    | -0.6109    | -0.684      | -1.1591     |
| OLS - factor model                | -0.5582    | -0.5564    | -0.8502    | -0.9428    | -1.1586     | -1.4772     |
| <b>SA with SA &amp; US Cycles</b> |            |            |            |            |             |             |
| OLS                               | -0.4230    | -0.3088    | -0.5485    | -0.4802    | -0.5163     | -0.7979     |
| Elastic Net                       | -0.3700    | -0.2765    | -0.5891    | -0.4128    | -0.4913     | -0.7848     |

# Classification

- Why focus on direction?
  - Predicted price change of 1 vs. realized change of 3 → No money lost, only missed opportunity
  - Predicted price change of -1 vs. realized change of 1 → Money lost due to incorrect direction
- Sign vs. size
  - Failing to predict the size → Opportunity cost
  - Failing to predict the sign → Actual financial loss

# Results

## Classification

Table 11: Regression from classification lens for US

|           | $rp^{(1)}$ | $rp^{(2)}$ | $rp^{(5)}$ | $rp^{(7)}$ | $rp^{(10)}$ | $rp^{(20)}$ |
|-----------|------------|------------|------------|------------|-------------|-------------|
| Accuracy  | 0.6095     | 0.6714     | 0.6452     | 0.6643     | 0.6571      | 0.6619      |
| Precision | 0.6650     | 0.7262     | 0.7486     | 0.7784     | 0.7931      | 0.8142      |
| Recall    | 0.5721     | 0.5936     | 0.5551     | 0.5732     | 0.5610      | 0.5798      |
| F1        | 0.6150     | 0.6533     | 0.6375     | 0.6602     | 0.6571      | 0.6773      |

# Results

## Classification

Table 12: Summary output of three models in US context

|           | OLS - full model | Elastic Net - full model | OLS - factor model |
|-----------|------------------|--------------------------|--------------------|
| Accuracy  | 0.6492           | 0.6516                   | 0.6452             |
| Precision | 0.7497           | 0.7543                   | 0.7648             |
| Recall    | 0.5726           | 0.5725                   | 0.5385             |
| F1        | 0.6486           | 0.6501                   | 0.6317             |

# Results

## Out-of-sample – Accuracy

**What proportion of the time did the model correctly predict the direction of returns?**

Table 13: Comparison of accuracy of models in US, SA and SA with SA & US cycles

|                                   | $rp^{(1)}$ | $rp^{(2)}$ | $rp^{(5)}$ | $rp^{(7)}$ | $rp^{(10)}$ | $rp^{(20)}$ |
|-----------------------------------|------------|------------|------------|------------|-------------|-------------|
| <b>US</b>                         |            |            |            |            |             |             |
| OLS - full model                  | 0.6214     | 0.6667     | 0.6476     | 0.6500     | 0.6452      | 0.6524      |
| Elastic Net - full model          | 0.6048     | 0.6619     | 0.6452     | 0.6643     | 0.6571      | 0.6595      |
| OLS - factor model                | 0.6024     | 0.6643     | 0.6405     | 0.6405     | 0.6429      | 0.6643      |
| <b>SA</b>                         |            |            |            |            |             |             |
| OLS - full model                  | 0.5093     | 0.4861     | 0.5046     | 0.4861     | 0.4861      | 0.4861      |
| Elastic Net - full model          | 0.4491     | 0.4306     | 0.5556     | 0.5000     | 0.5046      | 0.4861      |
| OLS - factor model                | 0.4537     | 0.5139     | 0.4769     | 0.4861     | 0.4676      | 0.4861      |
| <b>SA with SA &amp; US Cycles</b> |            |            |            |            |             |             |
| OLS                               | 0.6343     | 0.5880     | 0.6157     | 0.6435     | 0.6019      | 0.5741      |
| Elastic Net                       | 0.5880     | 0.5833     | 0.6435     | 0.6250     | 0.6296      | 0.5926      |

# Conclusion

- The factor model is highly effective in the US context
- Performance not as strong in SA
- Combined model improves results in the SA context
- Models are not entirely robust out-of-sample - negative  $R^2$ s
- Models produce good accuracy results

# Improvements

- Importance of market movements in classification
- Use of sample weights
- Objective function for the logistic regression:

$$- \left[ \sum_{i=1}^N w_i y_i (\beta_0 + x_i^T \beta) - \log (1 + \exp (\beta_0 + x_i^T \beta)) \right] + \lambda \left[ (1 - \alpha) \frac{\|\beta\|_2^2}{2} + \alpha \|\beta\|_1 \right]$$

where

$$* \quad w_i = \frac{|\text{return}_i|}{\sum_{i=1}^N |\text{return}_i|}$$

- Reduce dimensionality using selected principal components
- Incorporating non-Linear models

Thank you  
Enkosi  
Dankie